

Spreading 37 Points in the Plane

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Abstract

We place 37 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 33.16343+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 38-point configuration gives 35.26616; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 37 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 33.16343\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 37$ is not on the page. Deleting a point from our 38-point configuration gives 35.26616; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n37.txt
```

in <https://github.com/Ge-limin/math719>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (friedman-maxmin2d/).

A Exact value

```
numerator:
33163437477531118321228265
denominator:
999999999998302224836449
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

37 points, x and y per line. File: friedman-maxmin2d/coords/n37.txt.

```
-6.898532942832 6.443251489250 -5.080090872215 8.675767514991
-6.920894614890 7.975177176402 -4.134879728231 11.395590167789
-4.411595351792 6.919293027442 -3.261648801599 10.908283540733
-6.062259233111 8.487764028708 -5.393763712688 6.731289541158
-2.565722629585 7.272624004927 -4.766418031742 10.620245488824
-6.266994639321 7.218596168214 -3.197260933096 6.497279325963
-5.733990847785 9.432348523179 -5.408359257542 7.731183020520
-2.607748826029 10.151702532546 -5.038064675771 5.796688987373
-7.594459114845 10.078911025056 -2.252049789112 9.217101978760
-4.097922511319 8.863771001275 -3.224691584687 8.376464374219
-3.893187105110 10.132938861769 -5.122117068659 11.554846042610
-3.552959970013 7.431879879748 -4.055896314875 5.984692473656
-6.025302016199 5.955944862194 -3.239287129540 9.376357853581
-7.552432918401 7.199832497437 -2.237454244259 8.217208499398
-6.962920811334 10.854255704020 -4.426190896646 7.919186506804
-4.751822486889 9.620352009462 -7.908131955318 8.134433051223
-5.748586392638 10.432242002541 -6.104285429555 11.366842556326
-7.922727500172 9.134326530585 -6.935490159744 8.975070655764
-6.607221774417 9.919655150235
```